

Project Interim Report
ON
Warehouse Inventory Tracker (OOP Project)
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming With Java

SESSION: 2025-26



Submitted By

Student Name : Vaibhav Sharma

ERP ID : RU-25-11568

**Bachelor of Technology in Computer Science &
Engineering with AI in association with Google**

Under the Guidance of

Mr. Harsh Multani

**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING RUNGTA
INTERNATIONAL SKILLS UNIVERSITY BHILAI , CG**

Acknowledgment

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I am also thankful to my institution for providing me with the necessary resources and a learning environment to enhance my knowledge and skills. I would like to extend my appreciation to all the faculty members who have contributed directly or indirectly to my learning and development during this project.

Lastly, I would like to thank my friends and classmates for their support and cooperation. Their help and motivation played an important role in completing this project. This project has been a great learning experience, and I am grateful to everyone who contributed to its successful completion.

Vaibhav Sharma

Certificate

This is to certify that **Vaibhav Sharma** has successfully completed the project titled “**Warehouse Inventory Tracker**” as a part of academic requirements.

The work submitted is a bona fide record of the project carried out by the student under my guidance and supervision. To the best of my knowledge, this project work has not been submitted elsewhere for any other degree or qualification.

The project demonstrates the understanding and application of Object-Oriented Programming concepts and has been completed satisfactorily.

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Project Guide Signature:

Name of Guide: VAIBHAV SHARMA

Date:- 14-04-2026

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Sr.no

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ABSTRACT

The Warehouse Inventory Tracker is a software system designed to manage and monitor product stock efficiently in a warehouse environment. It focuses on maintaining accurate records of product details such as name, quantity, and minimum stock levels. By leveraging Object-Oriented Programming (OOP) concepts, the system ensures organized data handling and real-time visibility of inventory status. This helps prevent stock shortages and improves decision-making in inventory management processes.

The system is built using core OOP principles such as encapsulation, classes, and object management. A Product class is used to represent individual items, while an Inventory Manager class manages multiple products using arrays of objects. The use of this keyword ensures correct assignment of object attributes, making the code more structured and maintainable. Additionally, iteration and conditional logic are implemented to evaluate stock levels and perform necessary operations efficiently.

One of the key features of this system is its ability to detect low stock levels and generate alerts when the quantity of a product falls below a defined threshold. This simulates real-world warehouse operations and enhances proactive inventory control. Overall, the project provides practical exposure to designing modular and scalable systems while strengthening fundamental programming concepts, making it highly relevant for real-world applications and academic learning.

INTRODUCTION

The Warehouse Inventory Tracker is a system developed to efficiently manage and monitor inventory within a warehouse environment. In modern supply chains, maintaining accurate stock records is essential to ensure smooth operations, prevent shortages, and avoid overstocking. This project focuses on creating a structured and reliable solution that keeps track of product details such as name, quantity, and minimum stock levels, enabling better control over inventory flow.

The Warehouse Inventory Tracker also includes logic to monitor stock levels and generate alerts when inventory falls below a predefined threshold. This feature helps simulate real-world warehouse scenarios, where timely restocking is critical. Overall, the system provides a practical understanding of programming fundamentals while demonstrating how software solutions can be applied to solve real-life inventory management problems.

- Helps in efficient inventory tracking and management
- Reduces chances of stock shortages and overstocking
- Demonstrates real-world application of OOP concepts
- Improves code modularity and reusability
- Uses arrays of objects to handle multiple products
- Implements conditional logic for decision making

OBJECTIVE

The primary objective of the Warehouse Inventory Tracker is to design and implement a system that efficiently manages product inventory using Object-Oriented Programming principles. The system aims to store and organize product details such as name, quantity, and minimum stock level in a structured manner. By doing so, it ensures accurate tracking of inventory and reduces the chances of manual errors in stock management.

Another key objective of this project is to apply core OOP concepts such as encapsulation, classes, and object interaction in a practical scenario. The project emphasizes the use of `this` keyword for proper initialization of object attributes and demonstrates how multiple objects can be managed using arrays. This helps in building a modular and maintainable system, which can be easily extended in the future.

- To generate alerts when stock falls below the minimum threshold
- To apply iteration and conditional statements for inventory checking
- To build a system that simulates real-world warehouse operations
- To improve program structure, modularity, and code reusability
- To enhance problem-solving and logical thinking skills
- To create a base system that can be extended into advanced applications (GUI, database, web-based systems)

SCOPE

CURRENT SCOPE

The current scope of the Warehouse Inventory Tracker is focused on developing a basic system that efficiently manages and monitors product inventory using Object-Oriented Programming concepts. The system allows users to store product details such as name, quantity, and threshold, and perform operations like checking stock levels and identifying low-stock items. It provides a simple and effective way to simulate real-world inventory management using arrays, loops, and conditional logic. This makes it highly useful for learning purposes and small-scale applications where basic inventory tracking is required.

- Stores product details like name, quantity, and threshold
- Detects and displays low stock alerts
- Uses arrays of objects to manage multiple products
- Suitable for academic learning and small-scale systems

FUTURE SCOPE

In the future, the scope of this project can be expanded significantly by integrating advanced technologies and features. The system can be enhanced with a graphical user interface (GUI) or converted into a web or mobile application for better usability. It can also be connected to a database for persistent data storage and real-time updates. Additional features like automated stock updates, user authentication, reporting systems, and analytics can be implemented to make it suitable for large-scale industrial use.

- **Development of GUI using Java Swing / JavaFX**
- **Conversion into web or mobile application**
- **Implementation of user login and role-based access**
- **Adding real-time stock updates and notifications**
- **Generating reports and analytics dashboards**
- **Integration with barcode or QR code systems**
- **Deployment as a cloud-based inventory management system**

SYSTEM REQUIREMENT

HARDWARE REQUIREMENT

<i>Component</i>	<i>Minimum Requirement</i>	<i>Recommended Requirement</i>
<i>Processor</i>	Intel Core i3 or equivalent	Intel Core i5 or above
<i>RAM</i>	4 GB	8 GB or more
<i>Storage</i>	500 MB free space	1 GB or more
<i>Input Devices</i>	Keyboard	Keyboard & Mouse

SOFTWARE REQUIREMENT

<i>Software Component</i>	<i>Requirement / Version</i>
<i>Operating System</i>	Windows 7/10/11, Linux, or macOS
<i>Programming Language</i>	Java (JDK 8 or above)
<i>IDE / Editor</i>	VS Code / IntelliJ IDEA / Eclipse
<i>Compiler</i>	Java Compiler

METHODOLOGY

✓ Step 1

- Understand the problem
- Decide what system should do

✓ Step 2

- Create Product class
- Add name, quantity, threshold

✓ Step 3

- Use constructor with this keyword
- Set values properly

✓ Step 4

- Create Inventory Manager class
- Store products in array

✓ Step 5

- Use loop to check all products

✓ Step 6

- Check condition:
- If quantity < threshold → Low stock

✓ Step 7

- Show alert message for low stock

✓ Step 8

- Display all product details

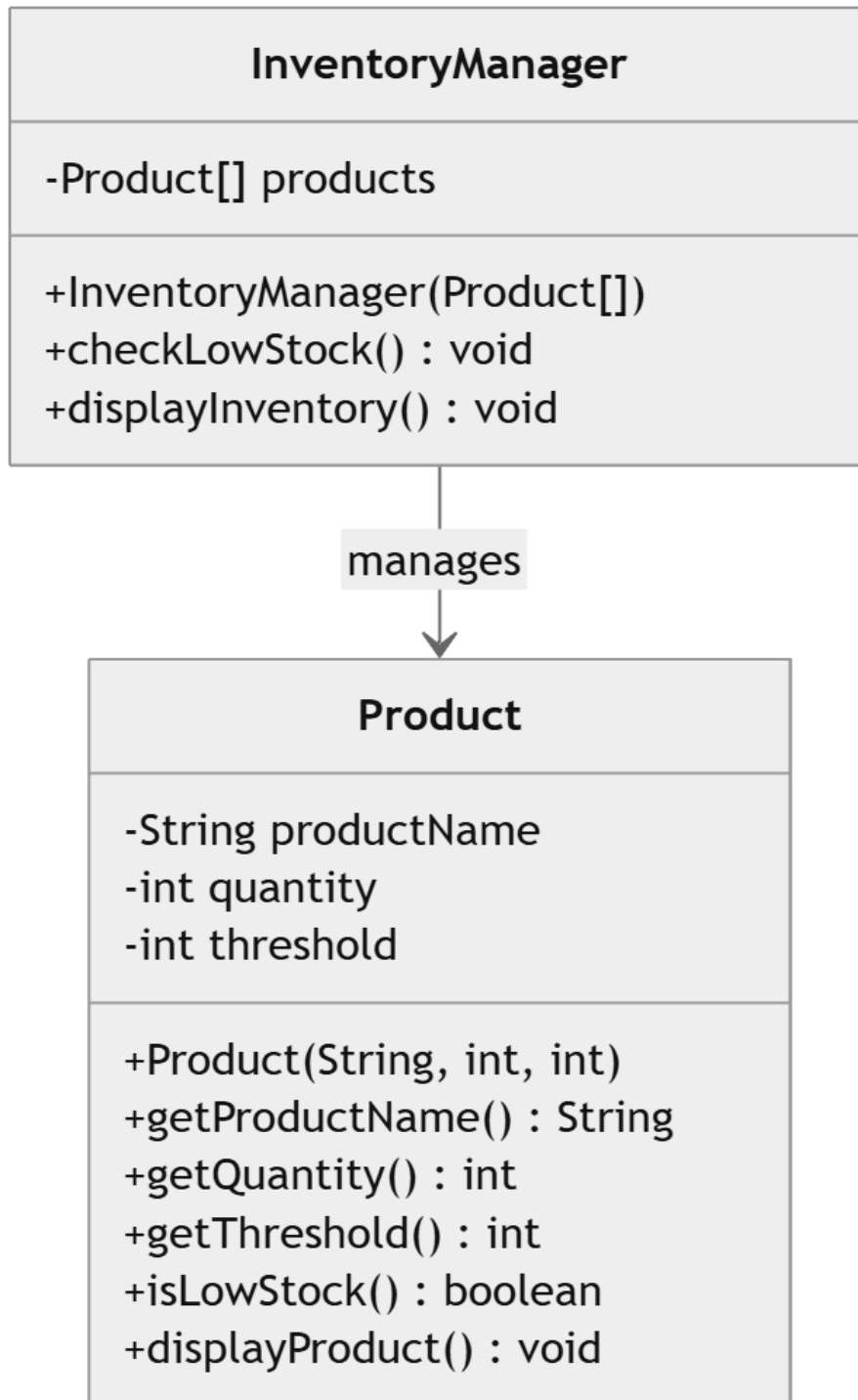
✓ Step 9

- Test everything in main method

✓ Step 10

- Check output and fix errors

CLASS DIAGRAM



IMPLEMENTATION

The implementation of the Warehouse Inventory Tracker is carried out using Java by applying Object-Oriented Programming principles. The system is divided into two main classes: Product and Inventory Manager. The Product class is responsible for representing individual items in the warehouse, while the Inventory Manager class handles the management of multiple products. This modular design ensures better organization and reusability of the code.

The Product class contains private attributes such as product name, quantity, and threshold to maintain encapsulation. A constructor is implemented using this keyword to initialize these attributes correctly. Getter methods are provided to access the values safely. Additionally, a method named is Low Stock () is used to check whether the quantity of a product is less than its threshold. Another method, display Product (), is used to print the details of each product along with its stock status.

- Created Product class with attributes and methods
- Used constructor with this keyword
- Applied encapsulation using private variables
- Created Inventory Manager class
- Stored products using array of objects
- Implemented low stock checking logic
- Used loops and conditions for processing
- Displayed output using methods

SAMPLE INPUT / OUTPUT

Study

WhatsApp

Inbox (662) - vaibha...

New Tab

Inbox (14) - vaibhav...

ChatGPT - Vaibhav

Rungta College Mail

Untitled notebook - ...

NotebookLM

Update

Programiz

Online Java Compiler

Programiz PRO

Main.java

Share

Run

Output

Clear

```
1 class Product {
2     private String productName;
3     private int quantity;
4     private int threshold;
5
6     public Product(String productName, int quantity, int threshold) {
7         this.productName = productName;
8         this.quantity = quantity;
9         this.threshold = threshold;
10    }
11
12    public String getProductName() {
13        return productName;
14    }
15
16    public int getQuantity() {
17        return quantity;
18    }
19
20    public int getThreshold() {
21        return threshold;
22    }
23
24    public boolean isLowStock() {
```

```
Laptop - Stock OK
ALERT: Mouse is low on stock!
Keyboard - Stock OK

Inventory Details:

Product: Laptop
Quantity: 10
Threshold: 5
Status: Stock OK
-----
Product: Mouse
Quantity: 2
Threshold: 5
Status: Low Stock
-----
Product: Keyboard
Quantity: 7
Threshold: 3
Status: Stock OK
-----

=== Code Execution Successful ===
```

FUTURE ENHANCEMENT

The Warehouse Inventory Tracker can be significantly enhanced by integrating a database system to enable permanent storage of inventory data. Currently, the system uses in-memory storage through arrays, which limits data persistence. By connecting the application to databases such as MySQL or MongoDB, users will be able to store, retrieve, and update inventory data even after the program is closed. This will make the system more reliable and suitable for real-world applications.

Another important enhancement is the development of a user-friendly interface, such as a graphical user interface (GUI) or a web-based application. This will allow users to interact with the system more easily without relying on command-line input. Features like forms for adding products, dashboards for viewing inventory status, and buttons for operations can greatly improve usability.

Additionally, implementing user authentication and role-based access control will make the system more secure and suitable for multiple users.

- Integration with database (MySQL / MongoDB)**
- Development of GUI using Java Swing or JavaFX**
- Conversion into web or mobile application**
- Implementation of user login and authentication system**
- Role-based access (Admin, Staff, Manager)**
- Real-time stock update and notifications**

- **Automated low stock alerts via email/SMS**
- **Addition of report generation and analytics dashboard**
- **Integration with barcode or QR code scanning**
- **Cloud deployment for remote access and scalability**
- **Support for bulk product import/export**
- **Performance optimization for large-scale inventory**

GANTT CHART

Project Timeline: Warehouse Inventory Tracker

Task / Phase	Week 1	Week 2	Week 3	Week 4
Requirement Analysis				
System Design (Classes)				
Coding – Product Class				
Coding – Inventory Manager				
Implementation of Logic				
Testing & Debugging				
Documentation & Report				
Final Review & Submission				

CONCLUSION

The Warehouse Inventory Tracker project successfully demonstrates how Object-Oriented Programming concepts can be applied to solve real-world problems. The system efficiently manages product details such as name, quantity, and threshold, and provides a mechanism to monitor stock levels. By organizing the system into classes like Product and Inventory Manager, the project ensures a modular and structured approach to inventory management.

Overall, the project serves as a strong foundation for developing more advanced inventory systems. It not only improves programming skills but also provides insight into how software solutions can optimize business processes. With further enhancements like database integration and user interfaces, this system can be expanded into a full-scale inventory management application.

- Successfully implemented a Warehouse Inventory Tracking System**
- Applied key OOP concepts (encapsulation, classes, objects)**
- Used this keyword for proper initialization**
- Managed multiple products using arrays of objects**
- Implemented low stock detection and alert system**
- Improved understanding of loops and conditional logic**
- Developed a modular and structured code design**
- Simulated real-world inventory management system**
- Built a strong base for future enhancements**
- Enhanced problem-solving and programming skills**

REFERENCES

Basic References

- Java Programming Documentation, Oracle
- Object-Oriented Programming Concepts – Study Material
- Class Notes and Lecture Materials
- Online Programming Tutorials

Detailed References

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<https://docs.oracle.com>
- Geeks for Geeks. *Object-Oriented Programming in Java*.
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- Study Material provided by college faculty

Personal implementation and coding practice

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